NARAYANA IIT - JEE ACADEMY - INDIA

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| Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-5(P1) | Date: 12-04-26 |
| Time: 3 Hrs | 2021_P1 | Max. Marks: 180 |

# KEY SHEET

## MATHEMATICS

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | **A** | 2 | **C** | 3 | **D** | 4 | **A** | 5 | **10** |
| 6 | **9** | 7 | **6** | 8 | **3** | 9 | **1** | 10 | **2** |
| 11 | **ACD** | 12 | **ABD** | 13 | **AB** | 14 | **ABD** | 15 | **AB** |
| 16 | **BD** | 17 | **7** | 18 | **2** | 19 | **9** | | |

## PHYSICS

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 20 | **C** | 21 | **A** | 22 | **D** | 23 | **B** | 24 | **1** |
| 25 | **5** | 26 | **0.40** | 27 | **0.17** | 28 | **2.80 to 2.83** | 29 | **26** |
| 30 | **AC** | 31 | **BCD** | 32 | **AD** | 33 | **BCD** | 34 | **AC** |
| 35 | **ACD** | 36 | **8** | 37 | **20** | 38 | **8** | | |

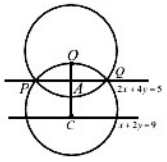
## CHEMISTRY

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 39 | **A** | 40 | **D** | 41 | **A** | 42 | **D** | 43 | **196** |
| 44 | **9** | 45 | **7.72** | 46 | **48.25** | 47 | **8.33** | 48 | **3** |
| 49 | **AC** | 50 | **AD** | 51 | **ACD** | 52 | **ABCD** | 53 | **ABCD** |
| 54 | **ABC** | 55 | **5** | 56 | **3** | 57 | **12** | | |

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# SOLUTIONS
## MATHS

1. $OA=\frac{\sqrt{5}}{2}, OC=\frac{4}{\sqrt{5}}$ $CQ=OC=\frac{4}{\sqrt{5}}$ and $CA=\frac{3}{2\sqrt{5}}$




$\therefore \quad OQ=\sqrt{OA^2+AQ^2}=\sqrt{OA^2+\left(CQ^2-CA\right)^2}$

$=\sqrt{\frac{5}{4}+\frac{16}{5}-\frac{9}{20}} \qquad =\sqrt{4}=2$

2. $a\left[2\sin\frac{3x}{2}\cos\frac{x}{2}\right]=2\sin\frac{3x}{2}\cos\frac{3x}{2}$

$\sin\frac{3x}{2}=0 \qquad a=4\cos^2\frac{x}{2}-3$

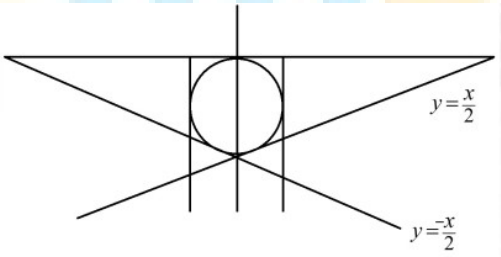
$\frac{3x}{2}=n\pi \qquad a\in(-3,1)$

$x=\frac{2x\pi}{3} \qquad a=-2,-1,0$

$x=\frac{2\pi}{3} \qquad$ But for $a=-2$

$x=\frac{2\pi}{3}$

3.




Centre $\left(0,\sqrt{5}\right)$

4. $15<x_1+x_2+x_3\le 20$

$x_1+x_2+x_3=16+r, r=0,1,2,3,4, x_i\ge 1$

$y_1+y_2+y_3=13+r \qquad [y_i=x_i-1]\ y_i\ge 0$

Now, the number of positive integral solution of

$y_1+y_2+y_3=13+r$ *is* ${}^{13+r+3-1}C_{13+r}$

i.e. ${}^{15+r}C_{13+r}={}^{15+r}C_2$

Thus, the total number of solution is

$\sum_{r=0}^{4} {}^{15+r}C_2={}^{15}C_2+{}^{16}C_2+{}^{18}C_2+{}^{19}C_2$

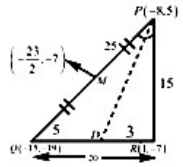
$=\frac{1}{2}(15\times14+16\times15+17\times16+18\times17+19\times18)=685$



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5.

Since the triangle is right – angled, the circumcenter is the midpoint of $PQ$ and the orthocenter is $R(1,-7)$. Hence, $RM=\left|\sqrt{\left(\frac{23}{2}+1\right)^2}\right|=12\frac{1}{2}$




6.

The coordinates of the incenter are given as follows:

$$\bar{x}=\frac{20(-8)+15(-15)+25(1)}{20+15+25}=-6$$

$$\bar{y}=\frac{20(5)+15(-19)+25(-7)}{60}=-6$$

Hence, incenter $I$ is $(-6,-6)$.

Now, the equation of side $PR$ is

$$y+7=\frac{12}{-9}(x-1)\text{ Or }4x-4=-3y-21\text{ Or }4x-3y+17=0$$

Inradius is given as the distance of $I$ fro side $PR$, i.e.,

$$\frac{|-24-18+17|}{5}=5$$

7.

$$V=\frac{1}{6}\left[\vec{a}\vec{b}\vec{c}\right]=\frac{1}{6}\sqrt{\begin{vmatrix}1 & 1/2 & 1/2\\ 1/2 & 1 & 1/2\\ 1/2 & 1/2 & 1\end{vmatrix}}=\frac{1}{6\sqrt{2}}$$

8.

The face DAB has normal along $\vec{a}\times\vec{b}$

The face DBC has normal along $\vec{b}\times\vec{c}$

If $\theta$ is the angle between the faces

$$\left|\vec{a}\times\vec{b}\right|\left|\vec{b}\times\vec{a}\right|\cos\theta=\left|\left(\vec{a}\times\vec{b}\right).\left(\vec{b}\times\vec{c}\right)\right|=\frac{\sqrt{3}}{2}.\frac{\sqrt{3}}{2}\cos\theta=\begin{vmatrix}\frac{1}{2} & \frac{1}{2}\\ 1 & \frac{1}{2}\end{vmatrix}=\frac{1}{4}$$

$$\Rightarrow\quad \cos\theta=\frac{1}{3}\Rightarrow\theta=\cos^{-1}\left(\frac{1}{3}\right)$$

9 & 10.

The given system is

$$\sin x\cos 2y=\left(a^2-1\right)^2+1,$$

And $\cos x\sin 2y=a+1$ ..... (i)

Since the L.H.S. of both the equations does not exceed 1, the given system may have solutions only for $a's$ such that

$\left(a^2-1\right)^2+1\le 1$ and $-1\le a+1\le 1$ ..... (ii)

$\left(a^2-1\right)^2+1\le 1$

Or $\left(a^2-1\right)^2\le 0$



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Or $\left(a^2-1\right)^2=0$

Or $a=1$

For $a=1$, equation $ocsx\sin 2y=a+1$ does not hold.

Thus, $a=-1$ only and we get

$\sin x\cos 2y=1$

$\cos x\sin 2y=0$ ..... (iii)

$\sin x\cos 2y=1$

$\Rightarrow \sin x=1,\cos 2y=1$

Or $\sin x=-1,\cos 2y=-1$

For which $\cos x\sin 2y=0$.

11. $2^{\frac{1}{5}}\left(\frac{8}{3}\right)\le\int_0^2 x^2\left(x^2+2\right)^{\frac{1}{3}}dx\le 6^{\frac{1}{5}}\times\frac{8}{3}$

12. $S=$ coefficient of $x^{10}$ in

$\left\{{}^{10}C_1\left(1+x\right)^{20}-{}^{10}C_1\left(1+x\right)^{18}+{}^{10}C_2\left(1+x\right)^{16}-{}^{10}C_3\left(1+x\right)^{14}+{}^{10}C_4\left(1+x\right)^{12}-{}^{10}C_5\left(1+x\right)^{10}\right\}$

$=$ coefficient of $x^{10}$ in ${}^{10}C_0\left[\left(1+x\right)^2\right]^{10}-{}^{10}C_1\left[\left(1+x\right)^2\right]^9+{}^{10}C_2\left[\left(1+x\right)^2\right]^8$

$-{}^{10}C_3\left[\left(1+x\right)^2\right]^7+{}^{10}C_4\left[\left(1+x\right)^2\right]^6+{}^{10}C_5\left[\left(1+x\right)^2\right]^5+{}^{10}C_6\left[\left(1+x\right)^2\right]^4+____$

$=$coefficient of $x^{10}$ in $\left[\left(1+x\right)^2-1\right]^{10}$

13. $p=\frac{d}{dx}\left[\left(x^3+1\right)g\left(x^3\right)\right]$

$p=\left(x^3+1\right)g'\left(x^3\right)3x^2+g\left(x^3\right)3x^2$

$p\left(7^{\frac{-2}{3}}\right)=24g'(7)+3g(7)$

None $g\left[f\left(x^5+x-2\right)\right]=g\left(x^3-1\right)$

$\therefore \quad g(7)=32,\quad g'(7)=\frac{27}{4}$

$\therefore \quad p\left(7^{\frac{-2}{3}}\right)=258$

Similarly $q=\frac{-3}{8}$.

14. Given, $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}=0$

$\Rightarrow \quad a^3+b^3+c^3-3abc=0$

$\Rightarrow \quad (a+b+c)\left(a^2+b^2+c^2-ab-bc-ca\right)=0$

$\Rightarrow \quad \frac{1}{2}(a+b+c)\left[(a-b)^2+(b-c)^2+(c-a)^2\right]=0$

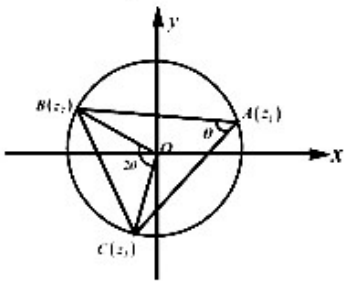


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$\Rightarrow \quad (a-b)^2+(b-c)^2+(c-a)^2=0$

$\Rightarrow \quad a=b=c\ [\because\ a+b+c\neq 0,\ \because\ z_1\neq 0,\ \because |z_1|=a\neq 0\ etc.]$

Hence, $OA=OB=OC$, where $O$ is the origin and $A,B,C$ are the points representing $z_1,z_2$ and $z_3$, respectively. Therefore, $O$ is cirucmcenter of $\Delta ABC$. Now,




$$\arg\left(\frac{z_3}{z_2}\right)=\angle BOC \qquad (1)$$

$$=2\angle BAC=2\arg\left(\frac{z_3-z_1}{z_2-z_1}\right) \qquad (2)$$

$$=\arg\left(\frac{z_3-z_1}{z_2-z_1}\right)^2 \qquad [\because\ \angle BOC=2\angle BAC]$$

Hence, $\arg\left(\frac{z_3}{z_2}\right)=\arg\left(\frac{z_3-z_1}{z_2-z_1}\right)^2$

Also, centroid is $(z_1+z_2+z_3)/3$. Since $HG:GO\equiv 2:1$ (where $H$ is orthocenter and G is centroid), then orthocenter is $z_1+z_2+z_3$ (by section formula). When triangle is equilateral centorid coincides with circumcenter; hence $z_1+z_2+z_3=0$.

Also, the area for equilateral triangle is $\left(\sqrt{3}/4\right)L^2$, where $L$ is length of side. Since radius is $|z_1|, L=\sqrt{3}|z_1|$, the area is $\left(3\sqrt{3}/4\right)|z_1|^2$.

15.

$$\int_0^{n+1} f(x)dx-\int_0^n f(x)dx=\int_n^{n+1} n^{x-n}dx=\frac{1}{n^n}\left[\frac{n^x}{\ln n}\right]_n^{n+1}=\frac{n-1}{\ln n}$$

$$\int_0^{n+1} g(x)dx-\int_0^n g(x)dx=\int_n^{n+1}(x-n)^n dx=\frac{(x-n)^{n+1}}{n+1}\Bigg]_n^{n+1}=\frac{1}{n+1}$$

$$t\in(0,1)\Rightarrow\int_0^t f(x)dx=0, t\in\left[1,\frac{\pi}{2}\right)\Rightarrow\int_0^t f(x)dx<\frac{\pi}{2}-1$$

$$t=1\Rightarrow\int_0^t g(x)dx=1$$

$$\lim_{n\to\infty}\int_0^n g(x)dx=\lim_{n\to\infty}\sum_{i=0}^{n}\frac{1}{i+1}\rightarrow \text{ Divergent series}$$

16.

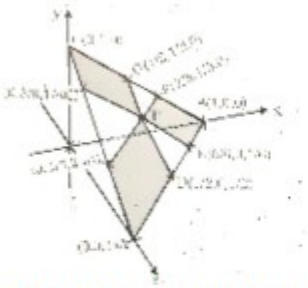
$A(\alpha)$ will be area of $\Delta ABC$ which is an equilateral

$\Delta A(\alpha)=\frac{\sqrt{3}}{4}(\sqrt{1+1})^2=\frac{\sqrt{3}}{2}$ sq. unit for $A(\beta)$, consider the point $P\left(\frac{1}{2},\frac{1}{3},\frac{1}{6}\right)$ in the plane $x+y+z=1$



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The region β is shown as shaded region



ΔPFG, PDE and PHI are equilateral triangles and we find required area by removing their total area from ar(ΔABC) Now FI is line through P parallel to AB

Equation of $FI$ : $\frac{x-\frac{1}{2}}{1}=\frac{y-\frac{1}{3}}{0}=\frac{z-\frac{1}{6}}{-1}$

$$F\left(\frac{2}{3},\frac{1}{3},0\right)$$

at $z=0;\left(0,\frac{1}{3},\frac{2}{3}\right)$

at $x=0;\left(0,\frac{1}{3},\frac{2}{3}\right)$

similarly we find points G, D and H, E

Required Area $=\frac{\sqrt{3}}{2}-\left[\frac{\sqrt{3}}{18}+\frac{\sqrt{3}}{72}+\frac{\sqrt{3}}{8}\right]$

17.

Since, the equation of asymptotes are

$3x-4y-1=0$

And $4x-3y-6=0$

$\therefore$ Equation of transverse axis is given by

$$\frac{|3x-4y-1|}{\sqrt{3^2+(-4)^2}}=\frac{|4x-3y-6|}{\sqrt{(4)^2+(-3)^2}}$$

Or $|3x-4y-1|=|4x-3y-6|$

$3x-4y-1=\pm(4x-3y-6)$

$\Rightarrow \quad x+y-5=0$

And $x-y-7=0$

$\therefore$ Transverse axis is given,

$ax+by-c=0 \; a,b,c\in N$

$\Rightarrow \quad ax+by-c=x+y-5=0$

$a=1,b=1,c=5$

18.

$\Rightarrow \quad a+b+c=7$

We have,

$P(x)=3x^3-9x^2+2x+5$

$P'(x)=9x^2-18x+2$

$P''(x)=18x-18$



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$$\Rightarrow \quad f(x)=\begin{cases}\dfrac{18x-18+18}{6}, & x\neq\dfrac{\pi}{2}\\ \sin^{-1}(ab)+\cos^{-1}(a+b-3ab), & x=\dfrac{\pi}{2}\end{cases}$$

$$\Rightarrow \quad f(x)=\begin{cases}3x, & x\neq\dfrac{\pi}{2}\\ \sin^{-1}(ab)+\cos^{-1}(a+b-3ab), & x=\dfrac{\pi}{2}\end{cases}$$

For continuity,

$$\sin^{-1}(ab)+\cos^{-1}(a+b-3ab)=\frac{3\pi}{2}$$

$$\Rightarrow \quad ab=1 \text{ and } (a+b-3ab)=-1\Rightarrow a+b=2$$

19. Let $\left(15+\sqrt{220}\right)^{19}=I+f, 0<f<1\Rightarrow\left(15-\sqrt{220}\right)^{19}=f_1, 0<f_1<1$

Adding both, $I+1=10\lambda$

$\left(15+\sqrt{220}\right)^{82}=I_1+f_2, 0<f_2<1\Rightarrow\left(15-\sqrt{220}\right)^{82}=f_3, 0<f_3<1$

Adding both, $I_1+1=10\lambda_1$

So, the value of $I+I_1$ will be of the type $10\lambda_3-2$

But, we also get $f+f_2>1$

## PHYSICS

20.

$\text{Tan}\,\theta=\frac{1}{4}$

$v_1=\frac{\pi}{3}\left[x_2^2.2R-x_1^2.R\right]=\frac{7\pi}{48}R^3$

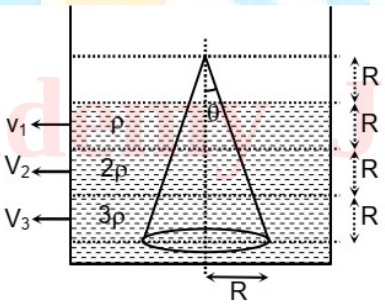
$v_2=\frac{\pi}{3}\left[x_3^2.3R-x_2^2.2R\right]=\frac{19\pi}{48}R^3$

$v_3=\frac{\pi}{3}\left[R^2.4R-x_3^2.3R\right]=\frac{37\pi}{48}R^3$

So, $m=v_1\rho+v_2(2\rho)+v_3(3\rho)$

$\rho_0\frac{\pi R^2}{3}\times 4R\left(\frac{3g}{2}\right)=\rho\frac{\pi R^3}{48}(156)\left(\frac{3g}{2}\right)$

Density of cone, $\rho_0=\frac{39}{16}\rho$




21. In equilibrium

$Mg = m\omega^2 r$

When the weight is displaced by a small distance x downwards and released.

From conservation of angular momentum

$m\omega r^2 = m\omega'(r-x)^2$

$T-Mg=Ma$

$$m(\omega')^2(r-x)-T=ma\Rightarrow m\omega^2 r\left(\frac{r^3}{(r-x)^3}\right)-Mg=ma$$

$$m\omega^2 r\left(1+\frac{3x}{r}\right)-Mg=(M+m)a\Rightarrow a=\frac{m\omega^2 r\left(\frac{3x}{r}\right)}{M+m}$$

Angular frequency $=\dfrac{3m\omega^2}{M+m}$ $\qquad \therefore f=\dfrac{1}{2\pi}\sqrt{\dfrac{3m\omega^2}{M+m}}$



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22. $f_P = \frac{3v_P}{4\ell_P},\ f_Q = \frac{3v_{mix}}{2\ell_Q}$

$$\frac{f_P}{f_Q} = \frac{\ell_Q}{2\ell_P}\frac{v_{H_2}}{v_{mix}} = \sqrt{\frac{\gamma_{H_2}}{\gamma_{mix}} \times \frac{T_{H_2}}{T_{mix}} \times \frac{M_{mix}}{M_{H_2}}} = \sqrt{\frac{7}{5} \times \frac{2}{3} \times \frac{400}{420} \times \frac{3}{2}} = \frac{2}{\sqrt{3}}$$

23. $6V_0 = -V_0 + 5V$

$$V = \frac{7}{5}V_0 = 1.4V_0 < \left(\sqrt{2}V_0 = V_e\right)$$

$\therefore$ Asteroid cannot escape and moves in an ellipse outside the original orbit.

24 & 25:

$v\cos 30° = \omega R\sqrt{3}\cos 60°$

$v = \omega R$ ...(i)

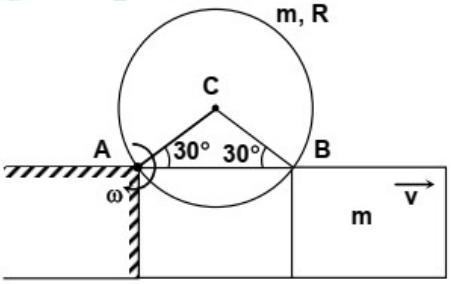
Now, using conservation of energy

$mgR(1 - \sin 30°) = \frac{1}{2}\times 2mR^2\omega^2 + \frac{1}{2}mv^2$

$\frac{mgR}{2} = \frac{3}{2}mv^2$

$\Rightarrow \quad gR = 3v^2$

$\Rightarrow \quad v = \sqrt{\frac{gR}{3}} = \sqrt{\frac{10\times 0.3}{3}} = 1\ m/s$




26. $Y = A\sin K \times \sin \omega t$ * Emf induced in adjacent

$V = A\omega \sin K \times \cos \omega t$ Loops will cancel out

$$Emf = \int Bv\ dx = BA\omega \cos \omega t \int \sin K \times dx$$

$$Emf_{max} = BA\omega\left[\frac{-\cos KX}{K}\right]_0^{\frac{\pi}{K}} \quad \text{for a single loop}$$

$$= \frac{2BA\omega}{K} = \frac{2BA\omega L}{5\pi} \quad K = \frac{2\pi}{\lambda} = \frac{2\pi}{2L}.5$$

$$\frac{5\lambda}{2} = L$$

27. $Z = R + \sqrt{3}Ri$

$$i = \frac{\varepsilon_0 \cos \omega t}{2R\angle 60^0} = \frac{\varepsilon_0}{2R}\cos\left(\omega t - \frac{\pi}{3}\right)$$

$$= \frac{BA\omega L}{5\pi R}\frac{\cos \pi}{6} \Rightarrow \beta = \frac{\sqrt{3}}{10} = 0.17$$

28. & 29.

$\tau = I\alpha$

$mgR - NR = 4mR^2\alpha$

$mg - N = 4m\alpha R$ ...(i)

$N = m\alpha R$ ...(ii)

From (i) and (ii)

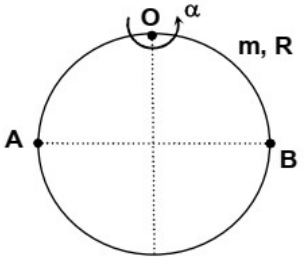
$mg = 5m\alpha R$

$\alpha = g/5R$

$\therefore$ acceleration of bead 'A' is

$a_A = \alpha R\sqrt{2} = \frac{g\sqrt{2}}{5} = 2\sqrt{2}\ m/s^2$

$\therefore$ acceleration of bead 'B' is $a_B = \frac{g}{5}\sqrt{26} = 2\sqrt{26}\ m/s^2$






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30.

$$\frac{m_0v^2}{r_n} = \frac{1}{4\pi\varepsilon_0}\frac{ze^2}{r_n^2} \qquad \text{(Where } m_0 = 50m\text{)} \quad \ldots(i)$$

$$m_0vr_n = \frac{n}{2}\frac{h}{\pi} \qquad \ldots(ii)$$

$$r_n = \frac{n^2h^2\varepsilon_0}{200\pi me^2}$$

$$E_n = \frac{-13.6z^2}{n^2} \times 50\ \text{eV}$$

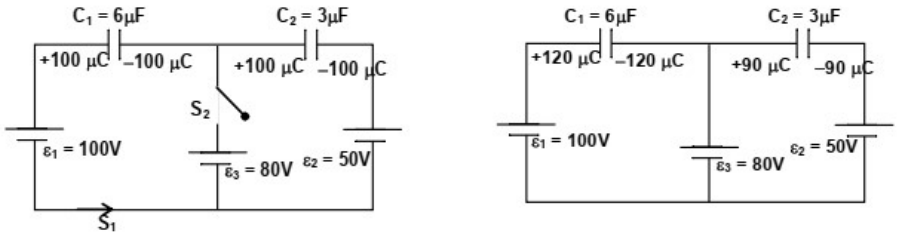
$E_4 = -680$ eV

$E_1 = -10880$ eV

$\Delta E = E_4 - E_1 = 10{,}200$ eV

$$\lambda = \frac{1240}{\Delta E} = 1.21\ \text{Å}$$

31.




The charge flown through the battery $\varepsilon_1$ and $\varepsilon_2$ after closing the switch '$S_2$',

$\Delta Q_1 = 120 - 100 = 20\ \mu C$

$\Delta Q_2 = -90 - (-100) = 10\ \mu C$

Now, $\Delta W_b = \Delta U + \Delta H$

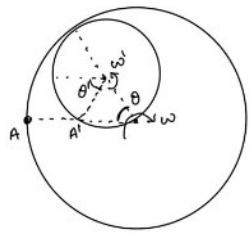
$20 \times 10^{-6} \times 100 + 10 \times 10^{-6} \times 50 - 30 \times 10^{-6} \times 80 =$

$$\frac{1}{2}\times 6\times 10^{-6}\times 400 + \frac{1}{2}\times 3\times 10^{-6}\times 900 - \frac{1}{2}\times 2\times 10^{-6}\times 2500 + \Delta H$$

$\Rightarrow 100 \times 10^{-6} = 50 \times 10^{-6} + \Delta H$

$\Rightarrow \Delta H = 50 \times 10^{-6} = 50\ \mu J$

32.




Rolling condition $\theta = \theta^1$ $\qquad \omega^1 R = u$

$$\omega R = u$$

$$\omega = \omega^1$$

$$X = -R\cos\theta - R\cos\theta^1$$

$$X = -2R\cos\omega t = -2Rco\left(\frac{u}{R}.t\right)$$

$Y_A = 0$ every point on the circumference of the disc will undergo SHM in a straight line (diameter of ring) with mean position at centre of the ring.

Point 'A' is confirmed to X-axis (SHM)

Point 'B' moves along Y=-X (SHM)

Point 'D' moves along Y = X (SHM)



For More Join: @IITJEE_Advanced

33.

$\Delta X_{max} = 2.5\,mm$

$n_{max} = \frac{2.5\times10^{-3}}{5000\times10^{-10}} = 5000$

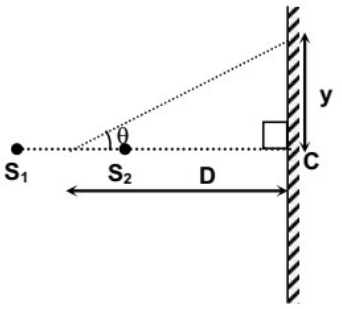
For 3000th order bright fringe, n = 3000

$d\cos\theta = (3000\,\lambda)$

$\cos\theta = \frac{3000\lambda}{5000\lambda} = \frac{3}{5}$

$\frac{D}{\sqrt{y^2+D^2}} = \frac{3}{5}$

Solving $y = \left(\frac{8}{3}\right)m$




For 5th bright fringe from 'C' n = 4996

$5000\,\lambda\cos\theta = 4996\,\lambda$

$\sin\theta = 0.04$

$\tan\theta = y/D = 0.04$

$y = 8\times10^{-2}\,m$

For 2500th order bright fringe, n = 2500

$\cos\theta = 1/2$

$\frac{D}{\sqrt{y^2+D^2}} = \frac{1}{2}$

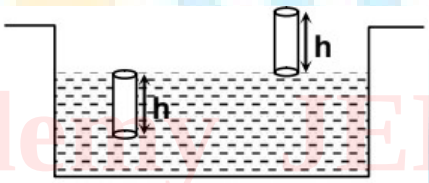
$4D^2 = y^2 + D^2$

$y = \sqrt{3}D$

$y = 2\sqrt{3}\ m$

35.

$(Ah)(\rho_\ell)\left[g\frac{h}{2}\right] = (Ah)(\rho_S g)\left(\frac{h}{2}+\frac{h}{2}\right)+\frac{1}{2}(Ah)(\rho_S)v^2$




$\frac{\rho_\ell gh}{2} = \rho_s gh + \frac{1}{2}\rho_s v^2$

$v = \sqrt{\frac{(\rho_\ell - 2\rho_s)gh}{\rho_s}}$

36. $V_{I_1} = V_{I_2} = m^2 V_0 \Rightarrow m_1 = -m_2$

$f_{upper} = 60\,cm,\ f_{lower} = 30\,cm$

$\frac{60}{60-u} = \frac{-30}{30-u}$

$\therefore 60-u = -60+2u$

$u = 40\ cm$

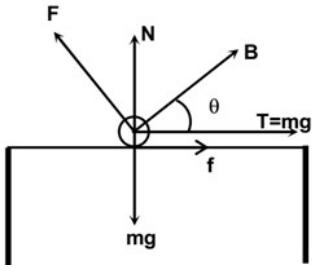
37. $A_{max} = 20,\ A_{min} = 2$

$\frac{I_{max}}{I_{min}} = 100 \Rightarrow 20\,dB$ difference

38.

$F_{min} = \frac{\mu mg + mg}{\sqrt{1+\mu^2}}$

$i\ell B_{min} = \frac{\mu mg + mg}{\sqrt{1+\mu^2}}$



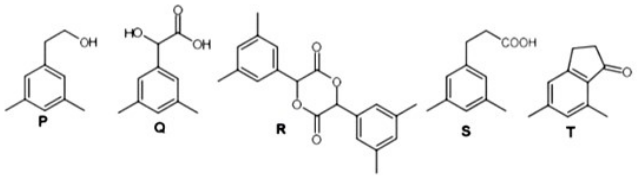



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# CHEMISTRY

39. Conceptual

40.

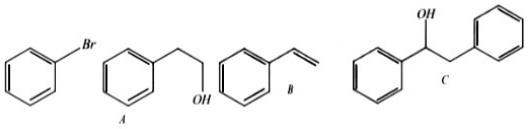


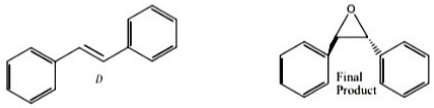

41. $\rho_{theo} = \dfrac{4\times 24}{\left(4\times 10^{-8}\right)^3 \times 6\times 10^{23}} = \dfrac{10}{4} g/cc$

$\%\ \text{occupancy} = \dfrac{2.4\times 10^3}{\frac{10}{4}\times 10^3}\times 100 = 96\%$

42. $A = \left[Cr(NH_3)_4 BrCl\right]Cl; B = \left[Cr(NH_3)_4 Cl_2\right]Br$

43 & 44:




45. By differentiating the polynomial w.r.t. T and then setting T = 298 K, we get temperature coefficient of cell.

From the data, n =1; $E^0 = +0.08V$

$\Delta_r G^0 = -nFE^0 = -1\times 96500\times 0.08\times 10^{-3} = -7.72\, kJ\, mol^{-1}$

46. $\left[\dfrac{\partial E^0}{\partial T}\right] = -5\times 10^{-4} VK^{-1} - 2\left(3.45\times 10^{-6}\right)(T-298)$

At T = 298K; $\left[\dfrac{\partial E^0}{\partial T}\right] = -5\times 10^{-4} VK^{-1}$

$\left[\dfrac{\partial E^0}{\partial T}\right] = \dfrac{\Delta S}{nF} \Rightarrow \Delta S = -5\times 10^{-4}\times 1\times 96500 = -48.25\, JK^{-1} mol^{-1}$

47. (i) At t=0, $0.2 = 2\times\dfrac{2}{M.W.}\times\dfrac{1000}{500} \Rightarrow M.W. = 40(Theoritical)$

(ii) After 30 minutes; $0.25 = 2\times\dfrac{2}{M.W.}\times\dfrac{1000}{500} \Rightarrow M.W. = 32(mixture)$

Degree of dissociation, $\alpha = \dfrac{40-32}{32(2-1)} = 0.25$

| | $A_{(aq)} \rightleftharpoons B_{(aq)}$ | + | $C_{(aq)}$ |
|---|---|---|---|
| $Initial$ | 0.1 | 0 | 0 |
| $At\ 30\ \text{min}$ | $(0.1-0.025)$ | 0.025 | 0.025 |

$K_{eq} = \dfrac{[B][C]}{[A]} = \dfrac{0.025\times 0.025}{(0.1-0.025)} = 8.33\times 10^{-3}$

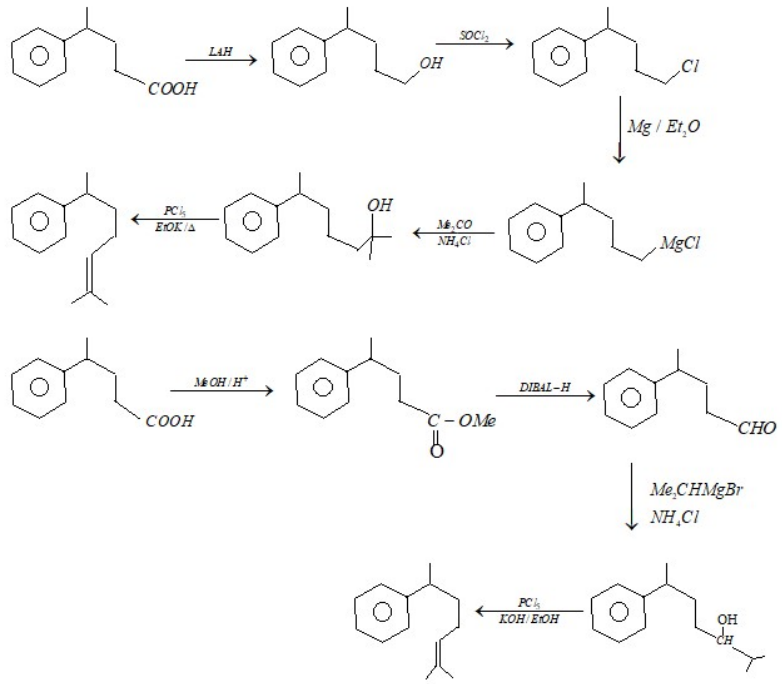
48. $i = \left[1+\alpha(n-1)\right] = 1+0.25(2-1) = 1.25$

$\pi = iCRT = 1.25\times 0.1\times 0.08\times 300 = 3.0\, atm.$

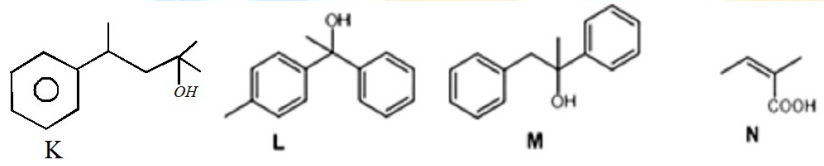


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49.




50.




51. A) For Tyndall effect there must be a large difference in refractive index between dispersed phase and dispersion medium in order to have diffraction of light.
C) Larger the coagulation value, lesser will be coagulating power.
D) Dissolving surfactant in a proper solvent will only form micelles at temperature above Kraft's temperature.

52. A) $K_a \times K_b = K_w$ (always) If $K_a = K_b \Rightarrow K_a = 10^{-7}$

B) $\text{pH} = \frac{1}{2}[7 - \log 0.1] = 4.0$ C) $\text{pOH} = \frac{1}{2}[7 - \log 0.1] = 4.0$

D) $[H^+]_{Total} = [H^+]_{HCl} = 0.01$ So, $\text{pH} = 2$

53. B) Since Alumina is covalent, to increase conductivity cryolite is added
D) In extraction of silver $Ag_2S$ is leached with KCN in presence of air:

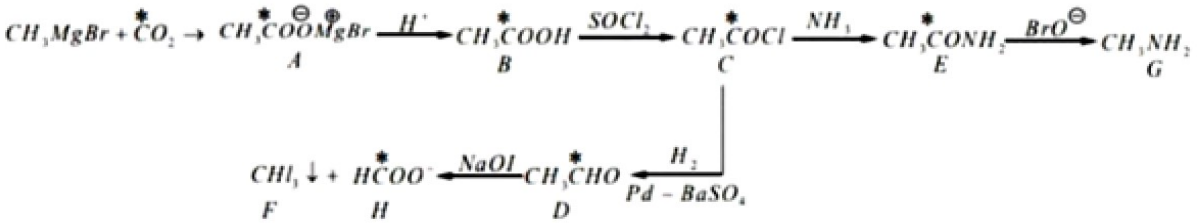
$$Ag_2S + NaCN + O_2 \rightleftharpoons Na\left[Ag(CN)_2\right] + Na_2S_2O_3$$

Thus, $O_2$ is oxidant.

$$2Ag(CN)_2^- + Zn \rightarrow \left[Zn(CN)_4\right]^{2-} + 2Ag \downarrow$$

55. With hemi acetal group carbohydrates acts as reducing sugar
For every one OH group (hydroxyl group) mass increases by 42 units during acetylation
$x = 7;\ y = 2$

56. Only I, III & VI are correct statements






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